

Ayeni Taiwo Michael

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🔍 Google Scholar | 📄 ResearchGate | 🌐 GitHub | 🔗 LinkedIn | 🌐 Website

Research Interests

- Survival Analysis and Lifetime Data Modeling
- Distribution Theory and Flexible Probability Models
- Empirical Likelihood Methods (JEL, AJEL, EJEL)
- Bayesian Inference and Simulation-Based Methods
- Reliability Modeling and Biomedical Applications

Education

Master of Applied Statistics (Biostatistics Concentration) Jan. 2026 – Present
Department of Mathematics & Statistics, Georgia State University, Atlanta, GA

- Advisor: [Prof. Yichuan Zhao](#)
- Thesis: *QTEG Distribution: Distribution Theory, Empirical Likelihood Inference, and Goodness-of-Fit Testing* (in progress)

Master of Professional Studies in Analytics (Statistical Modeling) Jan. 2024 – Jun. 2025
Northeastern University, Toronto, Canada

Master of Science in Statistics Apr. 2018 – Jan. 2021
Ekiti State University, Ado-Ekiti, Nigeria

- Thesis: *Determination of the Efficiency of the Exponential-Gamma Model and Formation of the Exponential-Gamma-X Family of Distributions*
- Supervisor: Prof. Olukunle Daniel Ogunwale

Bachelor of Science in Statistics Oct. 2010 – Jan. 2016
Ekiti State University, Ado-Ekiti, Nigeria

Research in Progress

Empirical Likelihood Inference for Skewed Lifetime Models: A Jackknife Approach for the QTEG Distribution (*In Progress*)

- Developing a jackknife empirical likelihood (JEL) framework for inference on QTEG parameters and survival probabilities
- Constructing an adjusted variant (AJEL) to improve finite-sample calibration in skewed lifetime settings
- Establishing Wilks-type chi-square limits for parameter inference, survival probabilities, and joint confidence regions
- Evaluating coverage performance against Wald-based intervals via Monte Carlo simulation

A Characterisation-Based Jackknife Empirical Likelihood Goodness-of-Fit Test for the QTEG Distribution (*In Preparation*)

- Developing a characterisation-based JEL goodness-of-fit test using a corrected symmetric kernel
- Constructing a Bartlett-corrected variant (BC-JEL) to improve finite-sample size calibration
- Targeting a Wilks-type χ^2 limiting distribution requiring no variance estimation or parametric bootstrap

Publications

Peer-Reviewed Journal Articles

- Agbona, A. A., Odukoya, E. A., Amalare, A., & **Ayeni, T. M.** (2025). Exponential-Gamma-Rayleigh Distribution: Theory and Properties. *Asian Journal of Probability and Statistics*, 27(3), 134–144. doi.org/10.9734/ajpas/2025/v27i3730
- Adeyiola, I. A., **Ayeni, T. M.**, & Odukoya, E. A. (2024). Sentiment Analysis Using Healthcare Data. *International Journal of Science and Research Archive*, 13(2), 3143–3148. doi.org/10.30574/ijrsra.2024.13.2.2476
- Faweya, O., Akinyemi, O., **Ayeni, T. M.**, & Tanimowo, M. O. (2023). Comparative Analysis on the Rate of Sexually Transmitted Diseases in Ekiti State. *International Journal of Research Publication and Reviews*, 4(4), 1693–1702. doi.org/10.55248/gengpi.2023.4.4.34559
- Faweya, O., Akinyemi, O., Ajayi, E. D., & **Ayeni, T. M.** (2023). Statistical Study of Rainfall Patterns in Ondo State, Nigeria. *International Journal of Novel Research in Physics Chemistry & Mathematics*, 10(2), 37–42. doi.org/10.5281/zenodo.8154921
- **Ayeni, T. M.**, Adeyiola, I. A., & Odukoya, E. A. (2023). Rayleigh-Exponential-Gamma Distribution: Theory and Properties. *International Journal of Research Publication and Reviews*, 4(8), 33–39. doi.org/10.55248/gengpi.4.823.48497

Manuscripts Under Review

- **Ayeni, T. M.**, Odukoya, E. A., & Zhao, Y. (2026). Quadratic Functional Variable Transformation of the Exponential-Gamma Distribution for Flexible Lifetime Modelling. *Statistical Papers*. Under review.

Manuscripts in Preparation

- **Ayeni, T. M.**, & Zhao, Y. Empirical Likelihood Inference for Skewed Lifetime Models: A Jackknife Approach for the QTREG Distribution. *Manuscript in preparation*.
- **Ayeni, T. M.**, & Zhao, Y. A Characterisation-Based Jackknife Empirical Likelihood Goodness-of-Fit Test for the Quadratic-Transformed Exponential-Gamma Distribution. *Manuscript in preparation*.

Research Experience

Graduate Research Assistant (GRA)

Jan. 2026 – Present

Department of Mathematics & Statistics, Georgia State University, Atlanta, GA

- Conducting methodological research under the supervision of **Prof. Yichuan Zhao**
- Developing the QTREG distribution and its empirical likelihood extensions for survival analysis
- Implementing JEL and AJEL inference frameworks and evaluating finite-sample performance via Monte Carlo simulation
- Developing characterisation-based goodness-of-fit tests using jackknife empirical likelihood methods

Master's Thesis Research

Apr. 2018 – Jan. 2021

Ekiti State University, Ado-Ekiti, Nigeria

- Developed the Exponential-Gamma-X (EG-X) family of distributions
- Derived theoretical properties and maximum likelihood estimation procedures
- Evaluated model performance through simulation studies and real datasets

Academic Appointments

Graduate Teaching Assistant (GTA)

Jan. 2026 – Present

Department of Mathematics & Statistics, Georgia State University, Atlanta, GA

- Provide academic support to undergraduate students enrolled in College Algebra
- Assist students with conceptual understanding, problem-solving strategies, and exam preparation
- Reinforce foundational algebraic reasoning and quantitative skills in a structured learning environment

University Teaching Experience

Assistant Lecturer

Mar. 2020 – Dec. 2023

Department of Statistics, Ekiti State University, Ado-Ekiti, Nigeria

- Taught probability theory, distribution theory, and statistical computing
- Supervised 40+ undergraduate research projects to completion
- Designed curriculum integrating theory with applied statistical methods
- Organised workshops on statistical computing and research methodology

Selected Projects

PredictRisk: Cardiovascular Risk Prediction (Bayesian Framework)

- Developed Bayesian models using PyMC and MCMC for six cardiovascular conditions
- Built an interactive Streamlit clinical decision-support application with uncertainty quantification

WorkWell AI: Burnout and Wellness Prediction System

- Developed machine learning models for behavioural prediction
- Built a full-stack application using Flask and React

Professional Experience

ESG Data Analyst Intern

Mar. 2025 – Jun. 2025

Re-Volt Charging Inc., Waltham, MA

- Applied statistical modelling to evaluate ESG impacts of EV charging infrastructure

Data Analyst Intern

Sep. 2024 – Dec. 2024

United Way Greater Toronto, Toronto, Canada

- Conducted NLP and statistical analysis of community socio-economic data

Presentations

- **Ayeni, T. M.**, Odukoya, E. A., & Zhao, Y. (2026). Quadratic Functional Variable Transformation of the Exponential-Gamma Distribution. *2026 NISS Virtual New Researchers Conference*, National Institute of Statistical Sciences. June 5, 2026. Oral Presentation.
- **Ayeni, T. M.**, Odukoya, E. A., & Zhao, Y. (2026). Quadratic Transformation of the Exponential-Gamma Distribution for Flexible Lifetime Modelling. *61st Annual SRCOS Summer Research Conference*, Southern Regional Council on Statistics, Pensacola Beach, FL. June 7–10, 2026. Poster Presentation. **Student Poster Competition Award (Cash Prize)**.
- **Ayeni, T. M.**, Odukoya, E. A., & Zhao, Y. (2026). Quadratic Transformation of the Exponential-Gamma Distribution for Flexible Lifetime Modelling. *Modeling the Invisible Workshop 2026*, Atlanta, GA. June 3–4, 2026. Poster Presentation.
- **Ayeni, T. M.**, Odukoya, E. A., & Zhao, Y. (2026). Quadratic Transformation of the Exponential-Gamma Distribution for Flexible Lifetime Modelling. *11th Workshop on Biostatistics and Bioinformatics*, ASA Georgia Chapter, Dean’s Meeting Center & College of Arts and Sciences Event Center, 25 Park Place, Georgia State University, Atlanta, GA. May 8–10, 2026. Poster Presentation. **Student Poster Competition Award (Cash Prize)**.
- **Ayeni, T. M.** (2026). The QTEG Distribution and Its Empirical Likelihood Extension for Survival Analysis. *End-of-Year Research Celebration, Department of Mathematics & Statistics, Georgia State University*, Atlanta, GA. April 10, 2026. Oral Presentation.
- **Ayeni, T. M.**, Odukoya, E. A., & Ilesanmi, A. O. (2025). Comparative Evaluation of Machine Learning Algorithms for Three-Class Diabetes Classification Using NHANES Data. *9th International Conference, Professional Statisticians’ Society of Nigeria (PSSN)*, Federal Polytechnic, Ado-Ekiti & Ekiti State University, Ado-Ekiti, Nigeria. August 4–7, 2025. Oral Presentation. In *Book of Abstracts: Advancing Statistical Literacy in AI Applications: Nigeria’s Strategy for Digital Transformation*.
- **Ayeni, T. M.** (2025). WorkWell AI: Predictive Wellness and Mental Health Monitoring. *Northeastern University Pitch Summit*. Research Demonstration.

Technical Skills

Statistical Methods:	Survival analysis, distribution theory, empirical likelihood, Bayesian inference, simulation
Programming:	R, Python, SQL, SAS
Tools & Frameworks:	PyMC, Stan/cmdstanr, ArviZ, L ^A T _E X, Overleaf, Streamlit
ML (Supporting):	XGBoost, Random Forest, PCA

Awards and Honors

- Student Poster Competition Award (Cash Prize), 61st Annual SRCOS Summer Research Conference 2026
- Student Poster Competition Award (Cash Prize), 11th Workshop on Biostatistics and Bioinformatics, ASA Georgia Chapter 2026
- Commonwealth Fellowship 2025
- Crest Africa Award for Best Innovator in Applied Statistics 2024
- Best Graduating Master’s Student, Ekiti State University 2021

Professional Memberships

- American Statistical Association (ASA) — Student Member
- Professional Statisticians’ Society of Nigeria (PSSN)
- Chartered Institute of Statisticians of Nigeria (CISON)

Certifications

- Using Data for Healthcare Improvement – Imperial College London
- Logistic Regression for Health Data – University of Michigan
- Linear Regression for Health Data – University of Michigan